

PRDA – 04

JOB MARKET ANALYSIS

PROJECT REPORT

Job Market Analysis

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Tool Used: Microsoft Power BI

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Page numbers refer to the final formatted report. The dashboard screenshot is included within Chapter 6.

Chapter 1 – Project Overview

This PRDA – 04 Job Market Analysis project analyzes 742 job market records containing 42 features and presents the results through a Microsoft Power BI dashboard. The analysis focuses on job concentration by state, salary patterns, industries, companies, job titles, education requirements, company characteristics, and other available features. The project brief requires the identification of states with the most jobs, salary comparisons, leading industries and companies, common job titles, skills required by companies, the relationship between salary and education, and multiple business insights.

The dataset contains salary fields in USD thousands (K). For example, a value of 101.48 represents approximately \$101,480. Revenue is stored as categorical ranges rather than a continuous numeric measure.

Power Query was used for data preparation and DAX measures were created for job count and salary KPIs. The final dashboard combines KPI cards with 12 analytical visualizations.

Chapter 2 – Project Objectives

- Obtain and prepare the job-market dataset for analysis in Power BI.
- Clean inconsistent values and represent unavailable information appropriately.
- Analyze states with the highest number of job postings.
- Compare average minimum, maximum, and average salaries across states.
- Identify the top industries and companies by job-posting volume.
- Identify the job titles with the largest number of job records and compare their salaries.
- Analyze the available skill-indicator fields and explain their intended interpretation.
- Examine salary patterns across education categories.
- Analyze company ownership, revenue groups, founding year, and company rating.
- Build a Power BI dashboard with KPIs and varied visualizations.
- Derive evidence-based findings and practical business suggestions.

Chapter 3 – Dataset Description

The PRDA – 04 project brief states that the dataset contains 742 rows and 42 features acquired from job-market sources. The fields cover job titles, salary information, company information, locations, industry, sector, revenue, company founding year, skills, seniority, and education indicators

Field	Description
Job Title	Original job title of the position.
Salary Estimate	Reported salary range.
Rating	Company rating.
Company Name	Name of the company.
Location / Headquarters	Job and company headquarters locations.
Founded	Year the company was founded.
Type of ownership	Ownership category such as private/public/government.
Industry / Sector	Business industry and sector.
Revenue	Company annual revenue range.
Lower / Upper Salary	Lower and upper salary values.
Avg Salary(K)	Average of lower and upper salary; K = USD 1,000.
Skill columns	Binary indicators: 1 required, 0 not required.
Jobtitle_sim	Standardized/similar job-title category.
seniority_by_title	Seniority extracted from the job title.
Degree	M or P where the job description gives degree-related experience credit.

The project documentation states that -1 means information was not available.

Chapter 4 – Data Connection and Preparation

4.1 Data loading

The job-market dataset was loaded into Power BI and prepared using Power Query before DAX calculations and visualization.

4.2 Empty-field removal

FIELD4 was removed because it contained no useful information.

4.3 Company name cleaning

The trailing company-rating text attached to New_Company_Name was removed so that company names could be used

consistently for grouping and distinct-company calculations.

4.4 Skill fields

Python, Spark, AWS, Excel, SAS, MATLAB, R, Power BI, Tableau, Azure, and SQL were converted to numeric whole-number indicators. A value of 1 means the skill is required and 0 means it is not required. The original -1 meaning of unavailable information was not globally converted to 0.

4.5 Missing values

For categorical fields, unavailable -1 values were represented as NA/Not Available. For numeric fields, -1 was converted to null. Genuine nulls were retained instead of inventing values through automatic imputation.

4.6 Location fields

Location and Headquarters were split into city and state components using text-before-delimiter and text-after-delimiter transformations.

4.7 Salary and revenue

Avg_SalaryK, Lower_Salary, and Upper_Salary were retained as numeric salary measures in USD thousands.

Revenue was retained as categorical ranges because the source data does not provide a continuous numeric revenue value.

Chapter 5 – KPI Analysis

The following KPI values are taken from the completed Power BI dashboard.

KPI	Value	Meaning
Total Job Postings	742	Total job records analyzed.
Companies Hiring	343	Distinct company names represented in the dataset.
Average Minimum Salary	74.75 USD K	Average lower end of reported salary ranges.
Average Salary	101.48 USD K	Overall average salary across the available Avg_SalaryK values.
Average Maximum Salary	128.21 USD K	Average upper end of reported salary ranges.

5.1 DAX Measures

Measure	DAX
Job Count	Job Count = COUNTROWS(Sheet1)
Average Salary	Average Salary = AVERAGE(Sheet1[Avg_SalaryK])
Average Lower Salary	Average Lower Salary = AVERAGE(Sheet1[Lower_Salary])
Average Upper Salary	Average Upper Salary = AVERAGE(Sheet1[Upper_Salary])
Companies Hiring	Companies Hiring = DISTINCTCOUNT(Sheet1[Company_Name])

5.2 KPI Interpretation

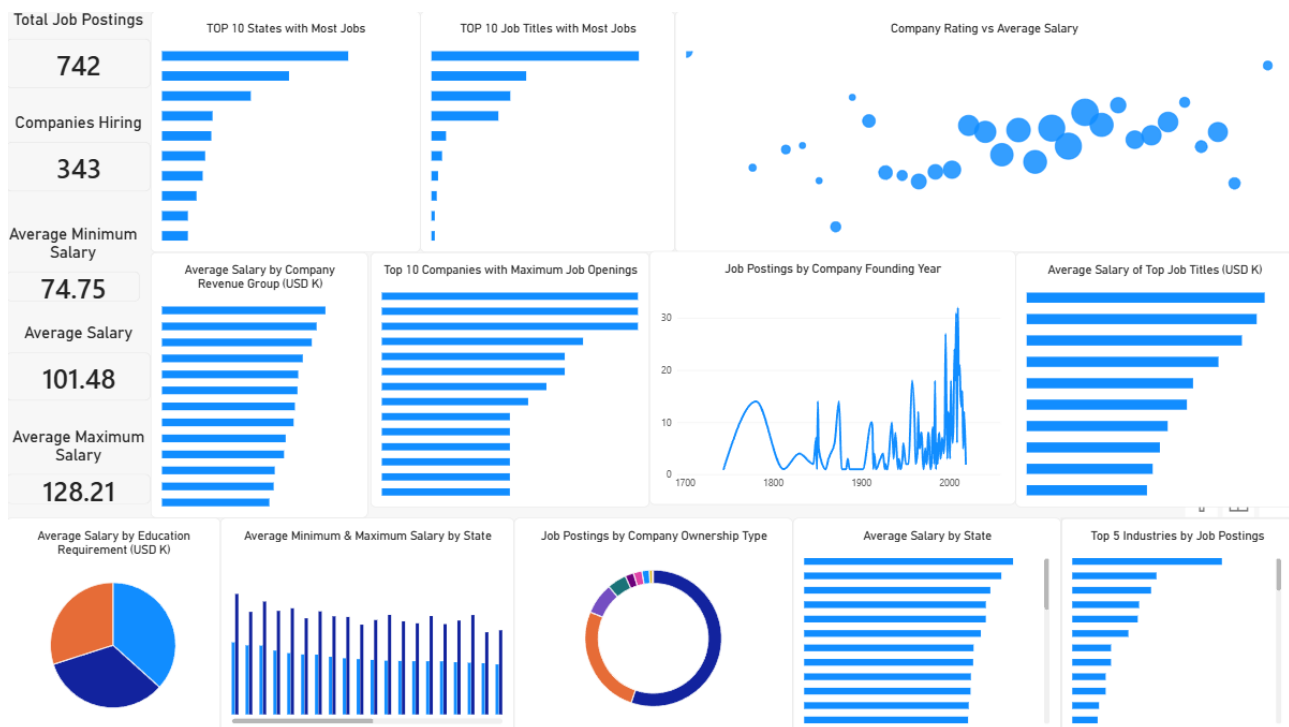
- The analysis covers 742 job records and 343 distinct companies.
- The overall average salary is 101.48K USD.
- The average lower and upper salary values are 74.75K and 128.21K USD respectively.
- The difference between the average lower and upper values indicates a substantial salary range across the analyzed job records.

Chapter 6 – Power BI Visualizations

The completed dashboard contains 12 visualizations in addition to the KPI cards. The visuals were selected to provide ranking, comparison, composition, distribution, and relationship views.

Visualization	Type	Purpose
Top 10 States with Most Jobs	Bar	Shows geographic concentration of job records.
Top 10 Job Titles with Most Jobs	Bar	Shows the most common standardized job-title categories.
Top 10 Companies with Maximum Job Openings	Bar	Shows companies with the largest number of job records.
Top 5 Industries by Job Postings	Bar	Identifies leading industries in the dataset.
Average Salary by State	Bar	Compares average salary across states.
Average Minimum & Maximum Salary by State	Clustered column	Compares lower and upper salary measures by state.
Average Salary of Top Job Titles	Bar	Compares salary levels among common job-title categories.
Average Salary by Education Requirement	Pie	Compares average salary across M, P, and NA categories.
Average Salary by Company Revenue Group	Bar/column	Compares average salary across categorical revenue groups.
Job Postings by Company Ownership Type	Donut	Shows the composition of job records by ownership type.
Company Rating vs Average Salary	Scatter	Shows the relationship pattern between rating and average salary.
Job Postings by Company Founding Year	Line	Shows job-record distribution according to company founding year.

Terminology note: the line chart uses Founded because the dataset does not contain a job-posting date. Therefore, it should be interpreted as job listings associated with companies by founding year, not as a temporal trend in posting activity.



Shows geographic concentration of job records.

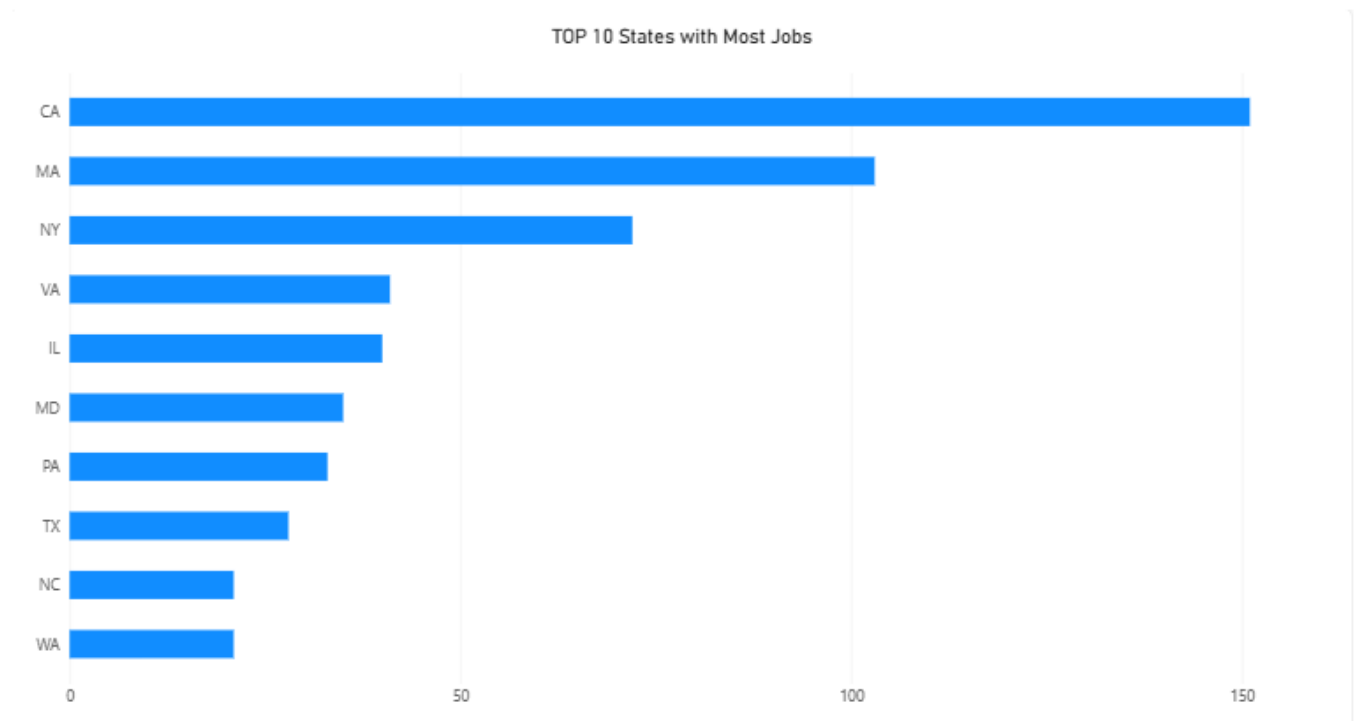


Fig 1

Shows the most common standardized job-title categories.

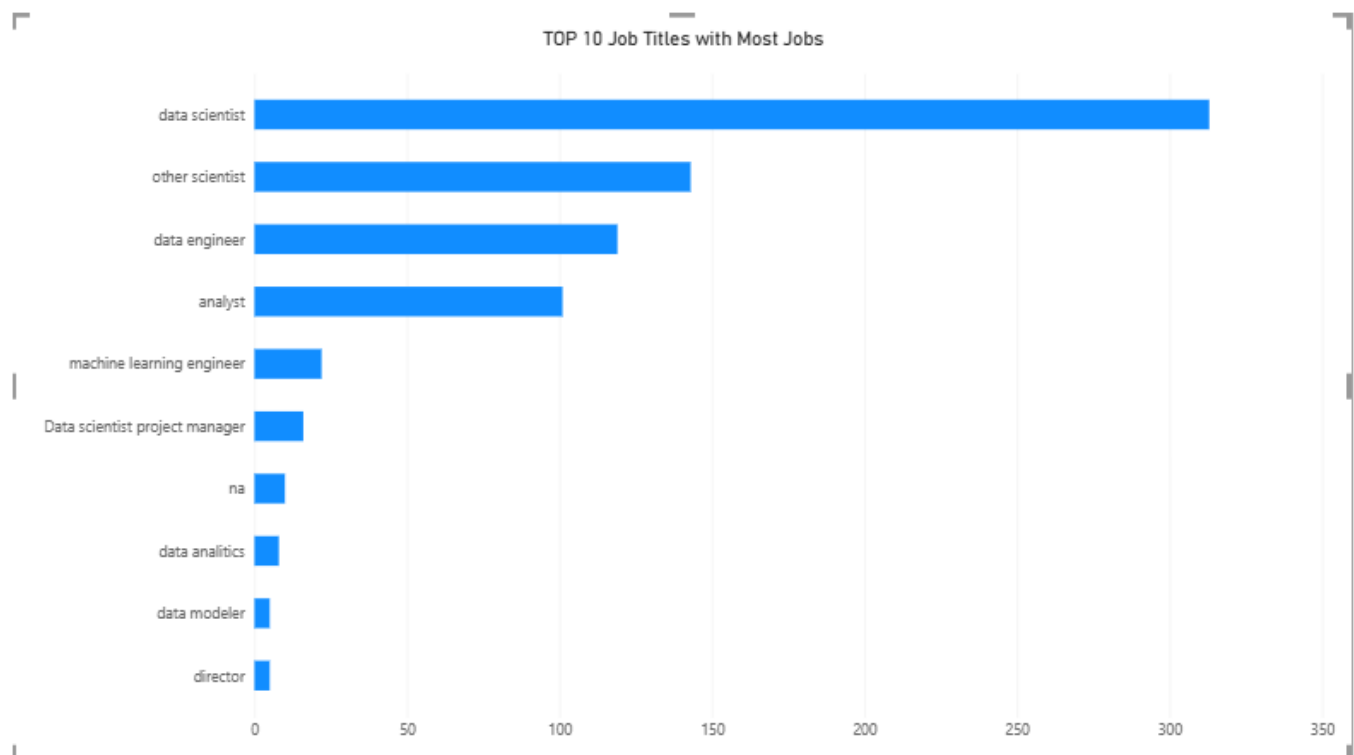


Fig 2

Shows companies with the largest number of job records.

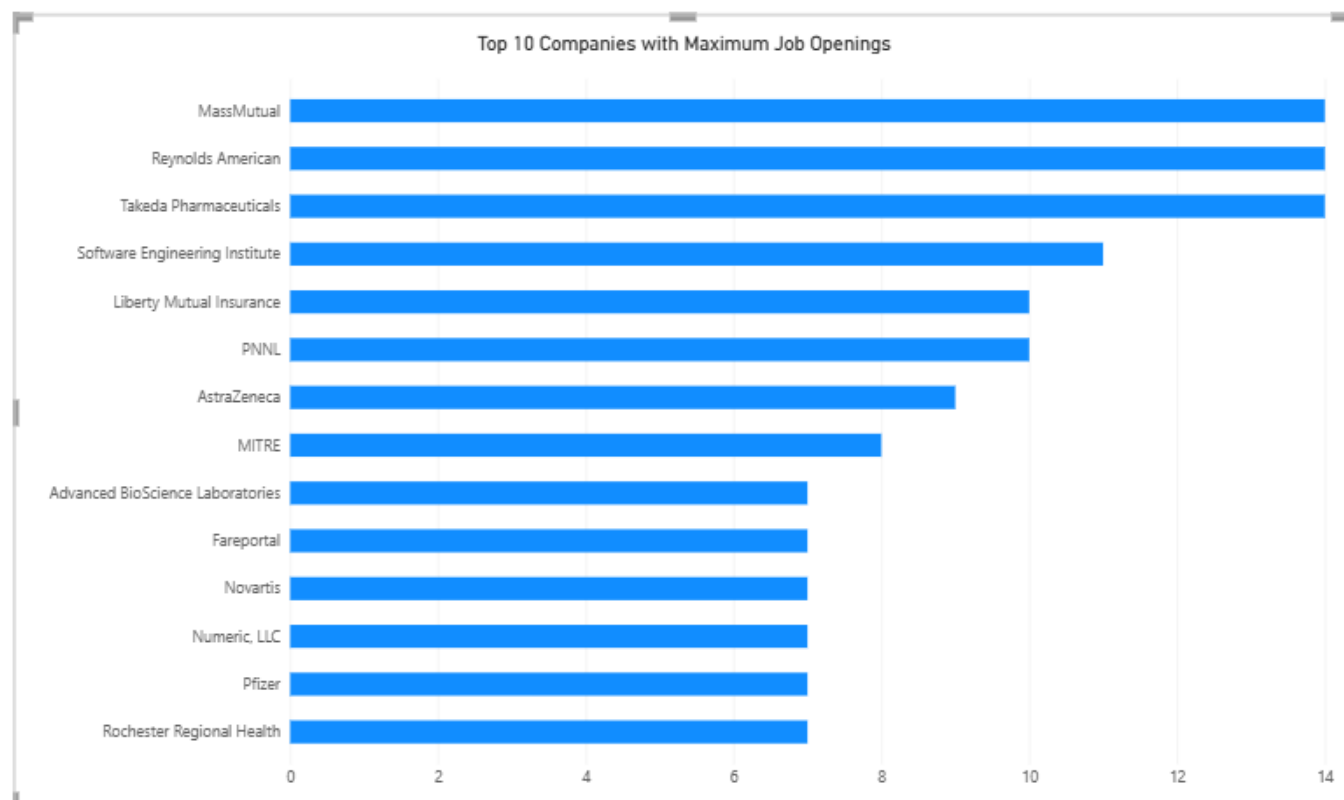


Fig 3

Identifies leading industries in the dataset.

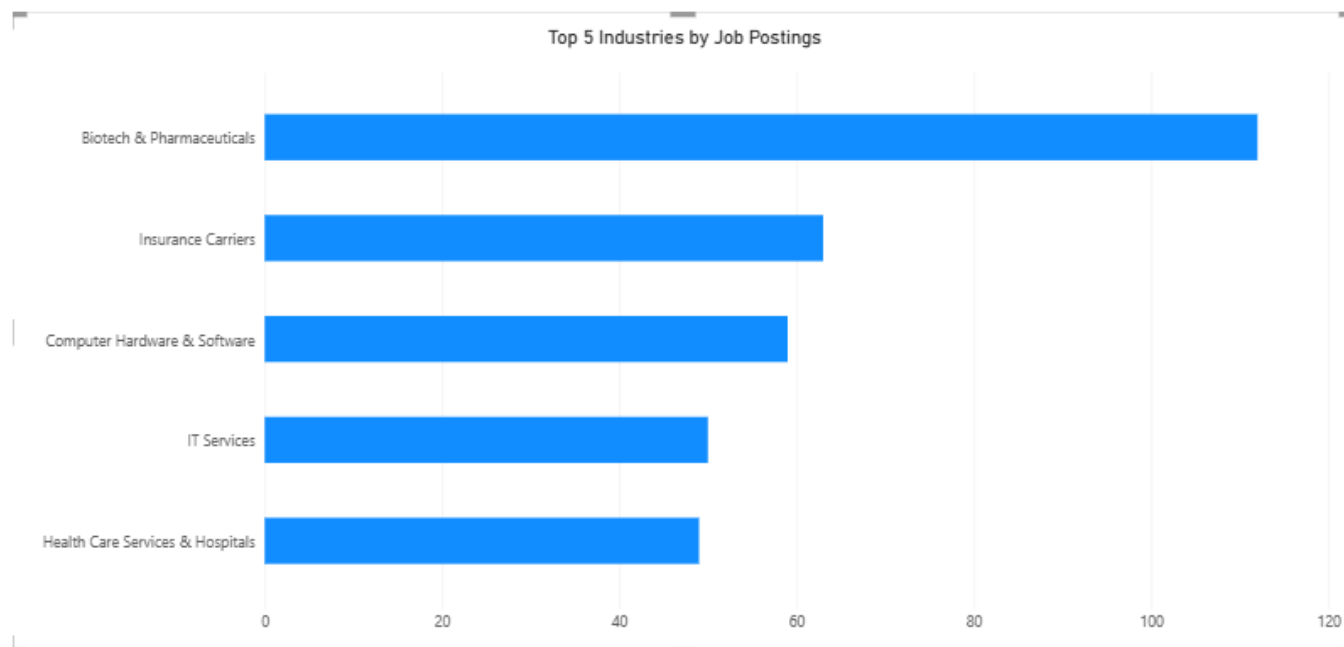


Fig 4

Compares average salary across states.

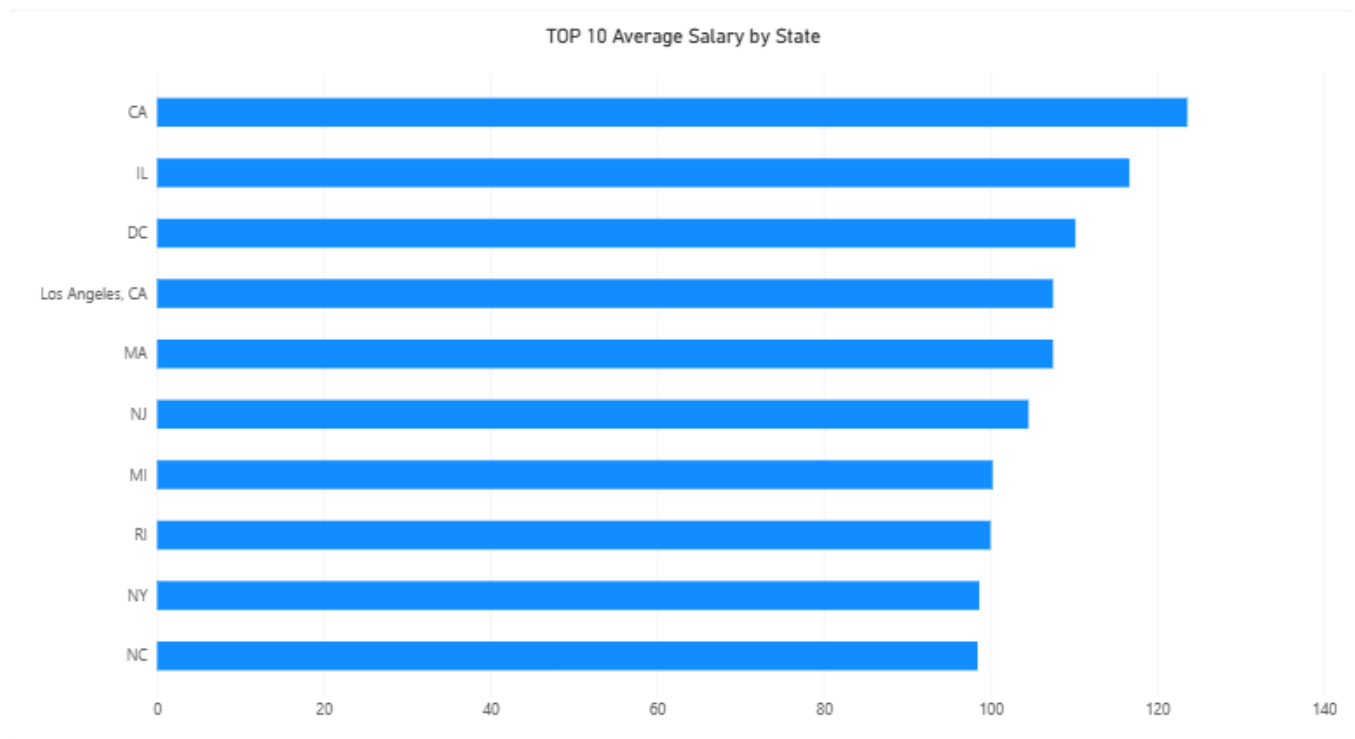


Fig 5

Compares lower and upper salary measures by state.

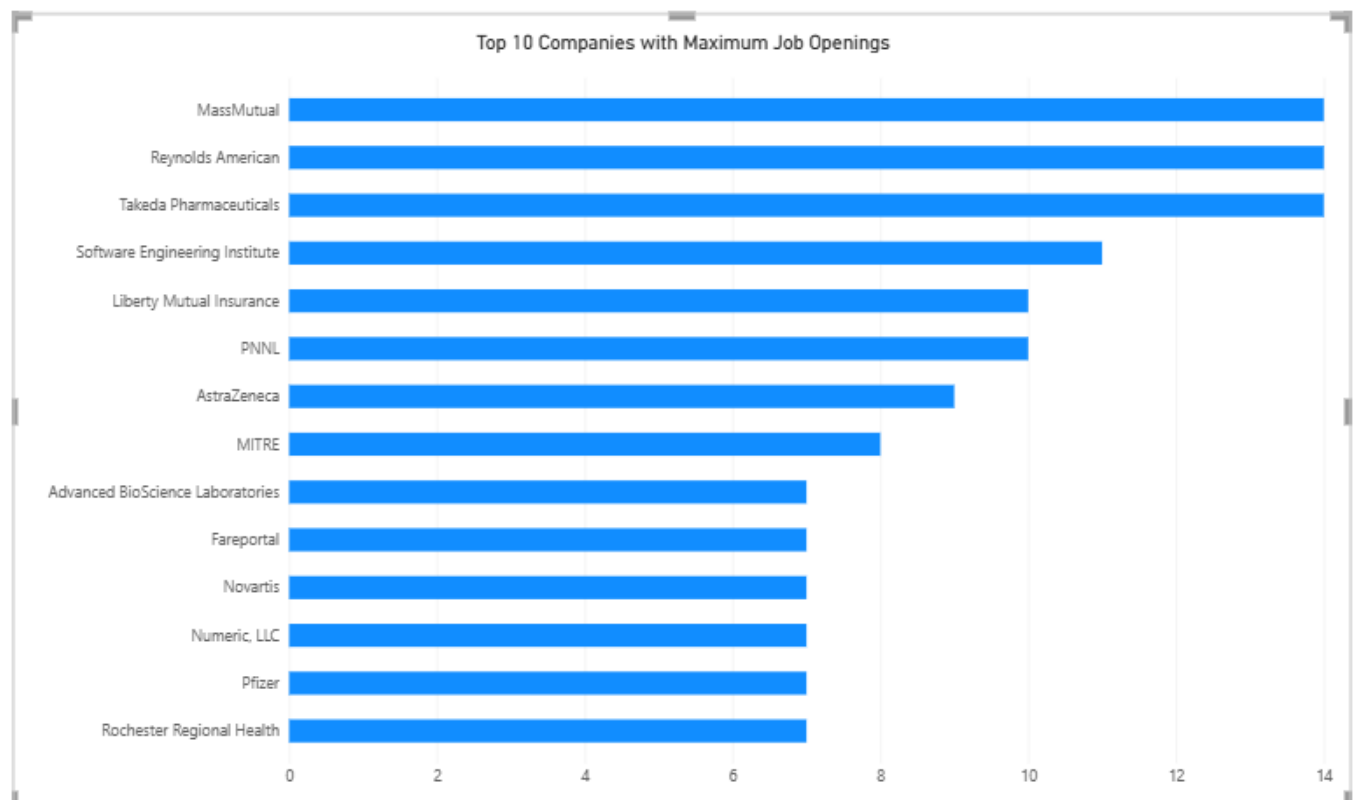


Fig 6

Compares salary levels among common job-title categories.

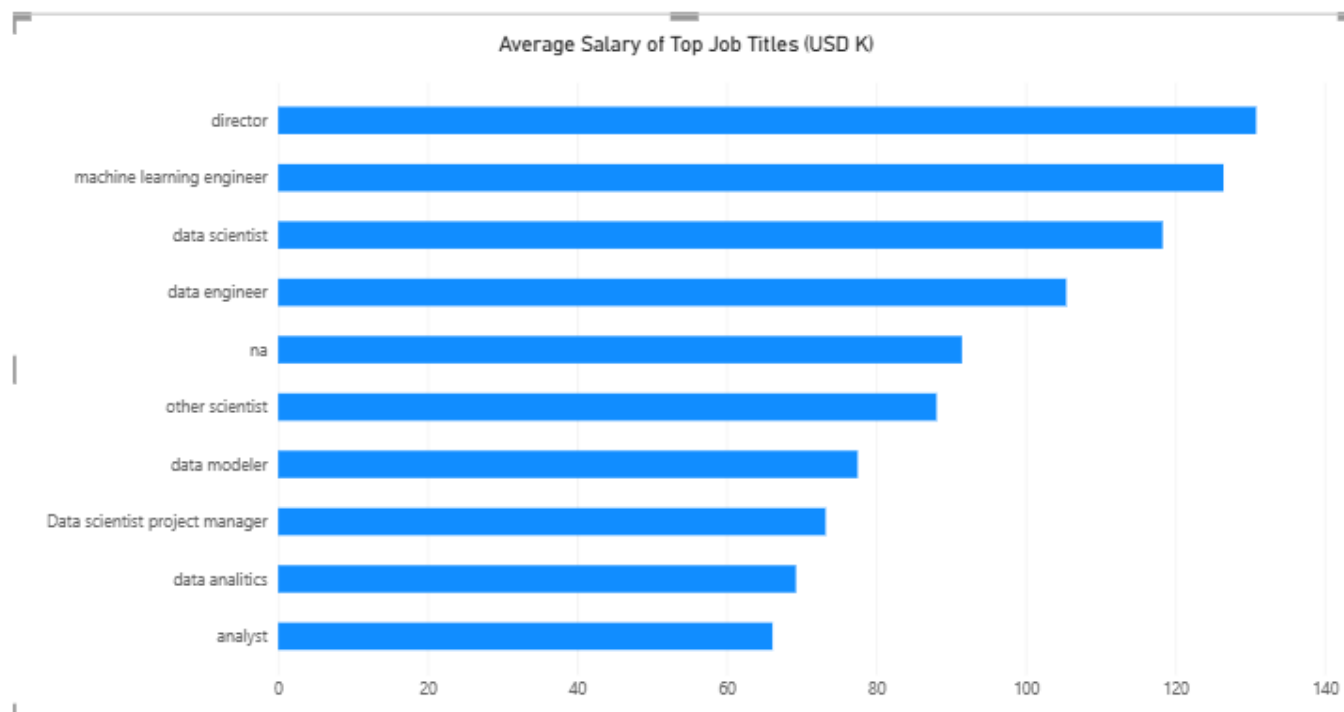


Fig 7

Compares average salary across M, P, and NA categories.

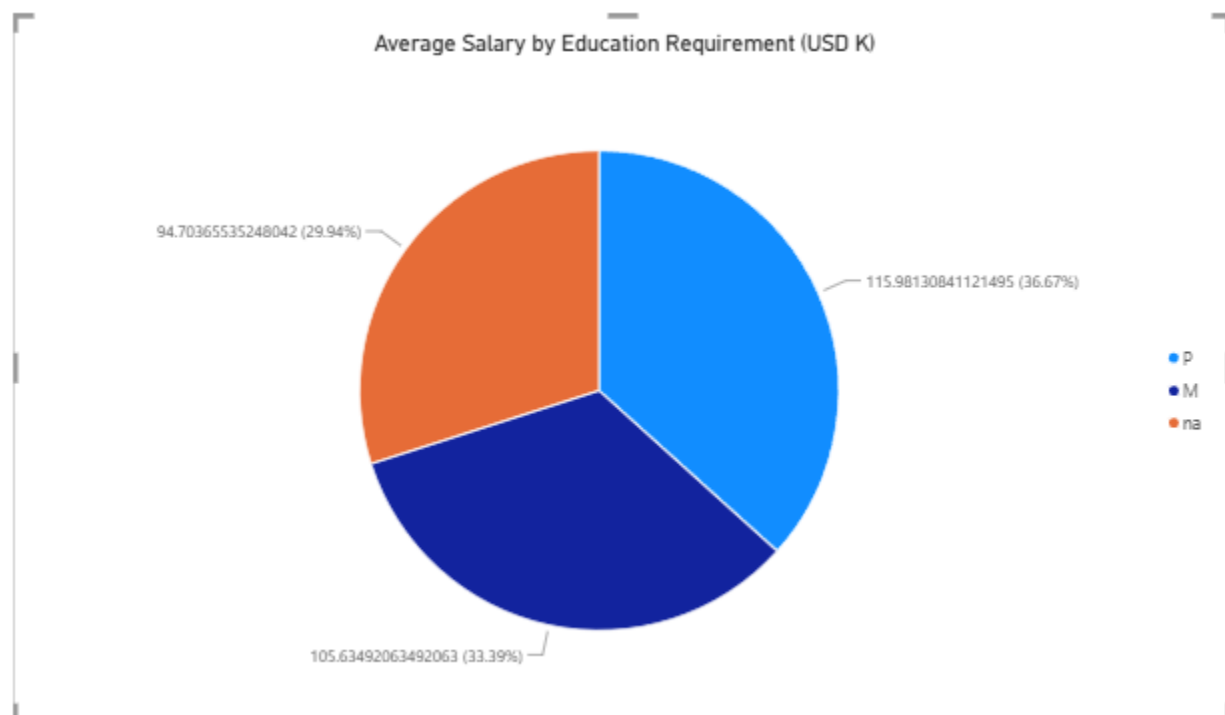


Fig 8

Compares average salary across categorical revenue groups.

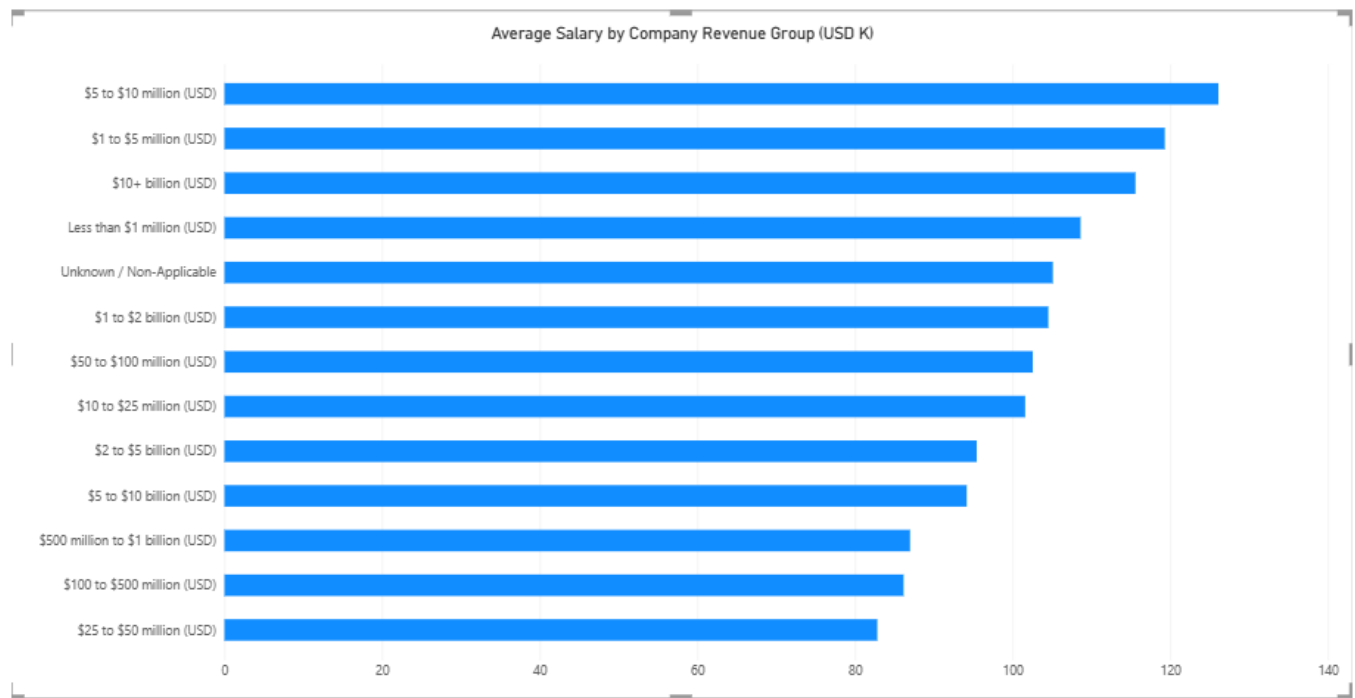


Fig 9

Shows the composition of job records by ownership type.

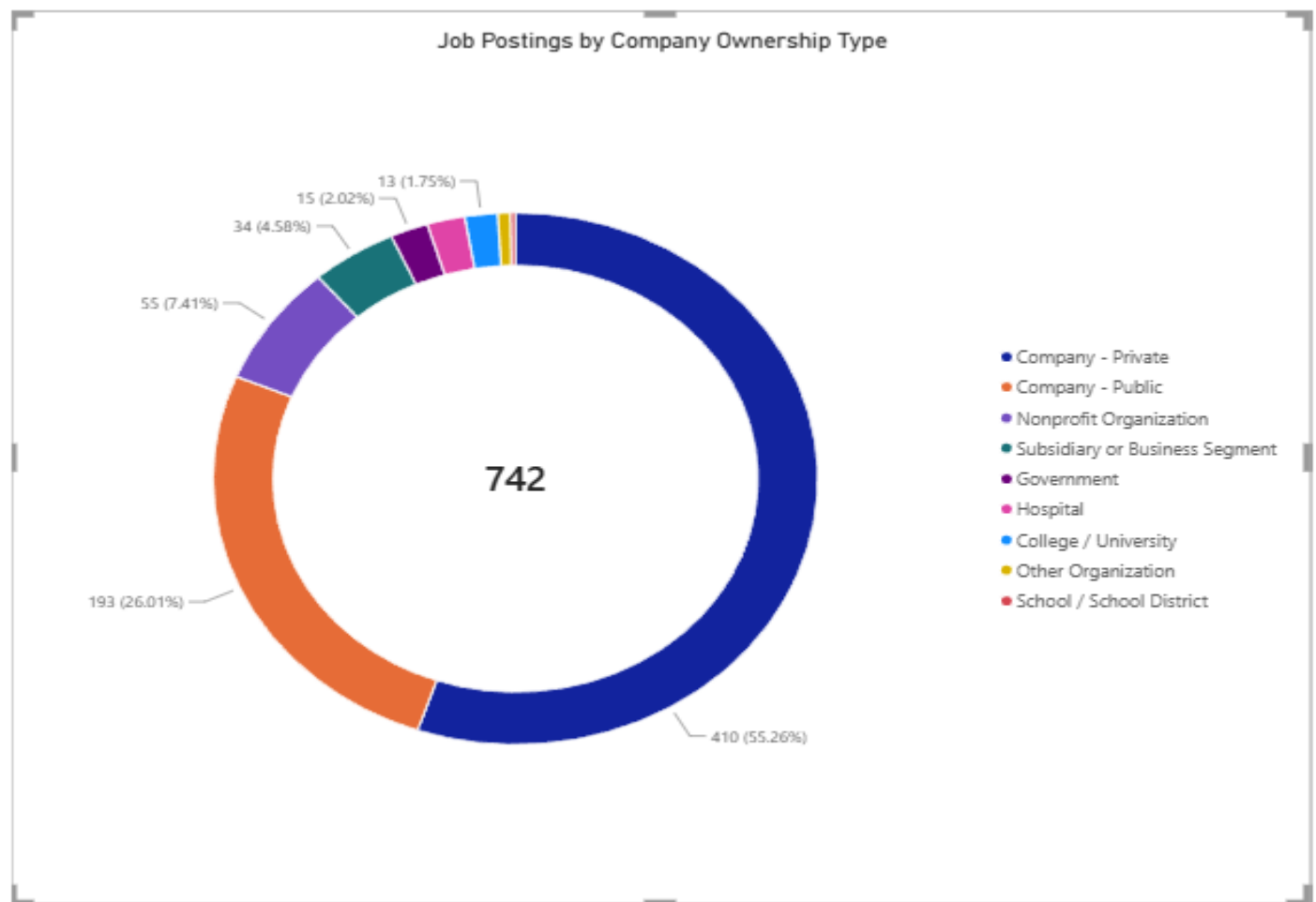


Fig 10

Shows the relationship pattern between rating and average salary.

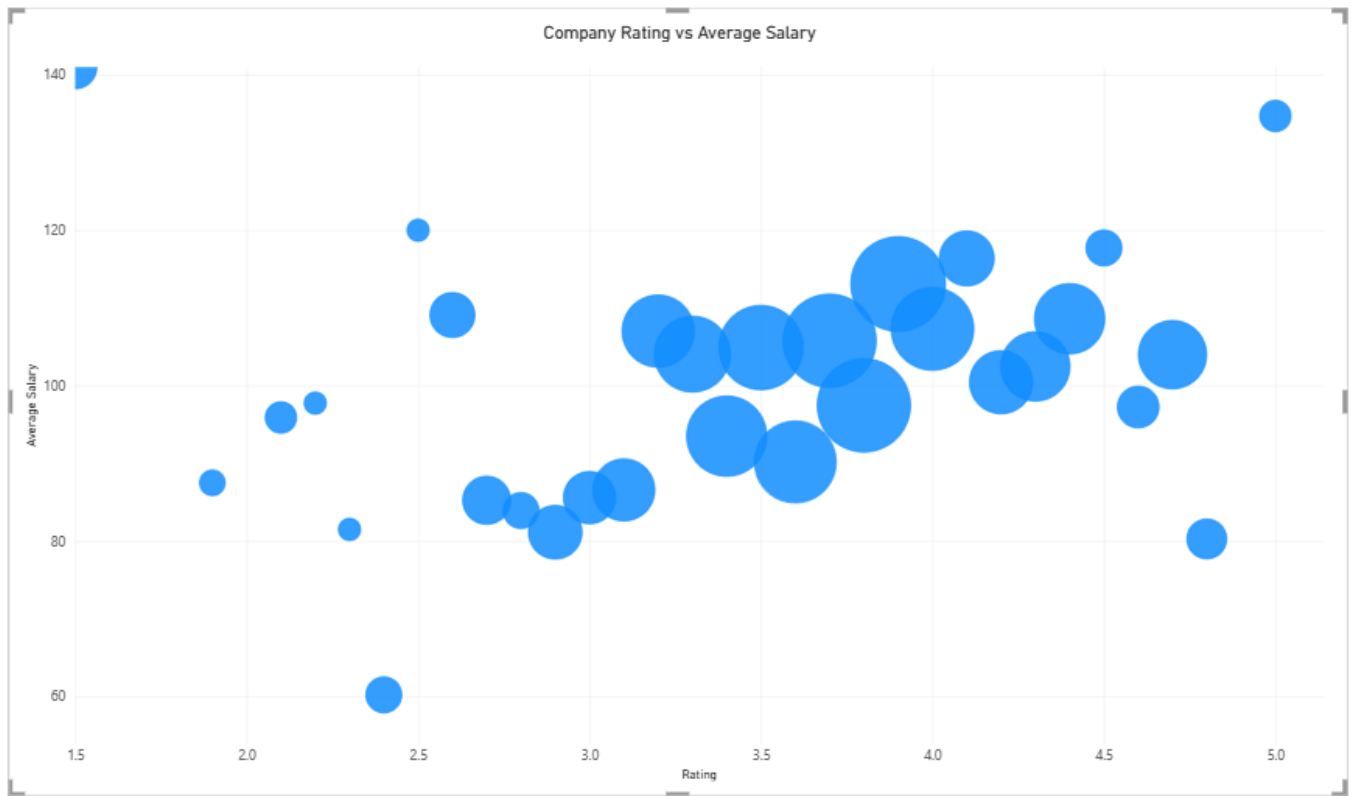


Fig 11

Shows job-record distribution according to company founding year.

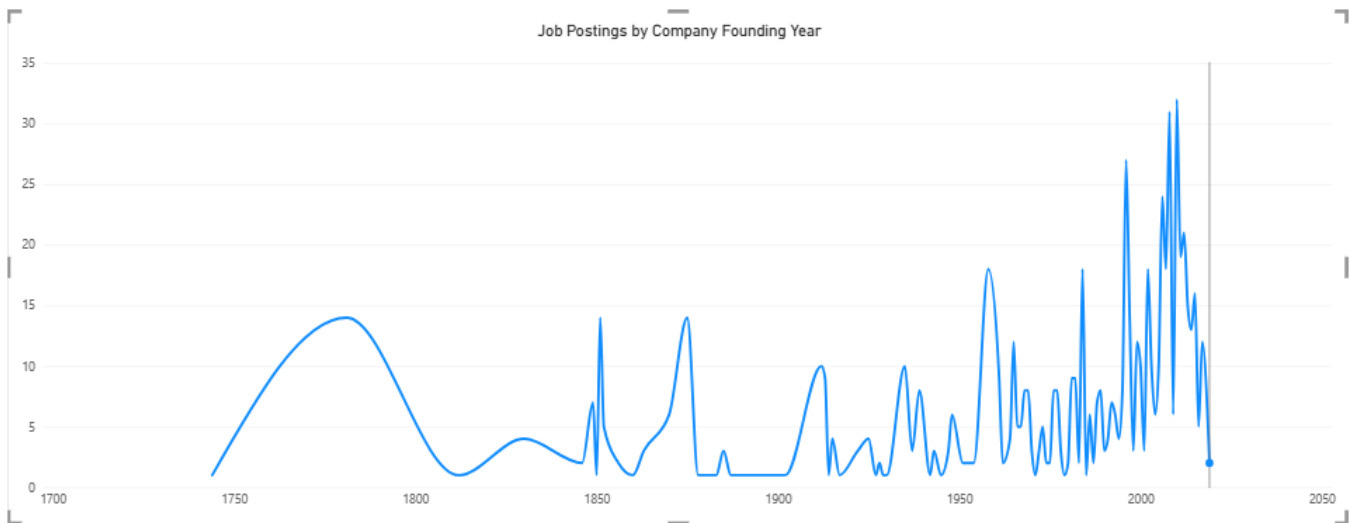


Fig 12

Chapter 7 – Key Findings

- Job-market size: The dataset contains 742 job records from 343 distinct companies.
- Salary level: The overall average salary is 101.48K USD, with average lower and upper salary values of 74.75K and 128.21K USD.
- Geographic concentration: California is the largest state category shown in the dashboard, followed by Massachusetts and New York among the displayed top states.
- Industry concentration: Biotech & Pharmaceuticals is the largest displayed industry by job postings, followed by Insurance Carriers and Computer Hardware & Software.
- Job-title concentration: Data Scientist is the largest displayed job-title category, followed by other scientist and data engineer categories.
- Company concentration: MassMutual, Reynolds American, and Takeda Pharmaceuticals appear among the highest-opening companies in the displayed ranking.
- Education and salary: The dashboard shows different average salary levels across P, M, and NA degree categories, with the P category visually highest among the three. This is an observed association and does not establish that the degree category causes higher salary.
- Revenue and salary: Average salary varies across the revenue groups shown. Because revenue is categorical, these results should be interpreted as group comparisons rather than a continuous correlation.
- Company rating and salary: The scatter chart shows variation in salary across company-rating values. The visual alone does not establish causation or a statistically significant relationship.
- Founding year distribution: Job records are concentrated more heavily among companies founded in later years in the displayed range, with several spikes. This describes the dataset composition rather than job-posting growth over time.

Chapter 8 – Business Suggestions

- Geographic targeting: Job seekers and employers can examine the states with the highest posting volumes when planning recruitment or market-entry activities.
- Industry targeting: Organizations involved in talent acquisition can prioritize industries with larger job-posting volumes when sourcing data-science talent.
- Role planning: Data Scientist, Data Engineer, Analyst, and related roles form important job-title categories in the dataset and can guide workforce planning and skill development.
- Salary benchmarking: The 74.75K average lower salary, 101.48K average salary, and 128.21K average upper salary provide a broad benchmark for the analyzed records, while state and title differences should be considered before making compensation decisions.
- Education planning: Employers can separately examine roles mentioning master's or PhD-related experience credit and compare those groups with other postings.
- Company segmentation: Revenue and ownership categories can be used to compare job-market patterns across different types of organizations.
- Skill planning: The binary skill fields can be used to identify which technical skills are most frequently requested overall and within individual job-title groups.
- Dashboard monitoring: Power BI can be reused as a management-reporting layer when new job-market data becomes available.

Chapter 10 – Conclusion

The PRDA – 04 Job Market Analysis project demonstrates an end-to-end Power BI workflow for analyzing job-market data. The dataset was cleaned and transformed in Power Query, DAX measures were created for the core KPIs, and the results were presented through a dashboard containing 12 analytical visualizations.

The analysis provides a view of job concentration by state, industry, company, and job title; salary levels across states, titles, education categories, and revenue groups; company ownership; company rating; and the distribution of job records across company founding years.

The dashboard converts the raw 742-row dataset into a structured analytical view that can support job-market exploration, compensation benchmarking, workforce planning, and further skill-demand analysis.

The results should be interpreted as descriptive findings from the supplied dataset and not as a complete representation of the broader job market.